Team ID: TM003665

SEVA FIRST INNOVATION CHALLENGE 2026

OFFICIAL SYNOPSIS • Education Department, Govt. of Gujarat • SSIP 2.0

PROBLEM STATEMENT CODE: PS0001

TRACK 2: BRING YOUR PROBLEM STATEMENT (OPEN INNOVATION)

Sankshep.ai: Automated GenAI Platform for Multi-Format Public Content Transformation

Mission:- An AI platform that turns ***one official document*** into summaries, advisories, posts, slides and video scripts in ***13 languages***, with a ***Private mode*** that keeps confidential files on the user's own device.

10 output formats | 13 languages | Private on-device mode | Live: sankshep-sfic.vercel.app

Sankshep.ai

One official document in.
Every citizen-ready format out.

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TRACK CATEGORY
College

INSTITUTION / ORGANIZATION
Vishwakarma Government Engineering College (VGEC), Ahmedabad, Gujarat

TEAM LEADER
Siddharth Devmurari

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PROBLEM STATEMENT & REAL-WORLD CONTEXT

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The Ground Challenge

Government departments, health systems and NGOs publish a constant stream of ***scheme guidelines, health advisories, circulars and safety alerts***. This information only helps when citizens can understand and act on it, yet turning it into usable formats is ***slow and manual.***

Root Cause:- one source document must become many audience-ready formats and languages ***by hand***, relying on scarce writing and translation skills, with no simple, safe, multi-format tool.

Repetitive Effort:- Same content rewritten for every audience and platform.

Inconsistency:- Different staff rewrite the same facts differently.

Language Barrier:- Official English vs. citizens who need Hindi & regional languages.

Skill Gaps:- Small offices & NGOs lack trained communication staff.

Delays:- Health, disaster & cyber-safety advisories lose value when late.

Data Privacy:- Confidential files can't be uploaded to public AI tools.

Target Beneficiaries & Flaws

Primary Users:-

- ❑ IEC & communication officers in departments and district administrations
- ❑ Health workers & PHC staff passing on advisories and scheme info
- ❑ Panchayat & municipal staff sharing notices and updates
- ❑ NGOs & community organisations explaining govt. schemes
- ❑ Disaster-management & cyber-safety teams issuing advisories

End Beneficiaries:- Citizens, especially those who don't read English or find official language hard to follow.

Secondary:- Departments with confidential documents that can't use outside AI, served by ***Private mode.***

Existing Gap

- ❑ Commercial AI writing tools produce ***one output at a time.***
- ❑ They need ***prompting skill*** that field staff rarely have.
- ❑ Documents go to ***outside servers***, not acceptable for confidential government material.
- ❑ Translation, slides and social posts each need a ***separate tool.***

Evidence *(proof for the pain points above)*

- ❑ **Language barrier:-** India has ***22 official languages*** (Eighth Schedule), yet most govt. content is in English, so many citizens can't follow it.
- ❑ **Repetitive effort:-** one circular needs ***5 hand-written versions*** (summary, explainer, post, slides, video script).
- ❑ **Delays:-** health, disaster & cyber advisories ***lose value*** when they arrive late.

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PROPOSED SOLUTION & INNOVATION NOVELTY

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HIGH-LEVEL EXECUTIVE SUMMARY

Sankshep.ai turns ***one official document*** (PDF, Word, scanned page, web link or text) into ***up to 10 citizen-ready formats at once***, in 13 languages, with an on-device ***Private mode*** so confidential files never leave the user's computer.

How Sankshep.ai Works

Upload
PDF, Word, image, link or text

→

Understand
Text extraction, OCR & key facts

→

Configure
Formats, audience, tone, language

→

Transform
All drafts generated in parallel

→

Review & Refine
Edit in plain words or regenerate

→

Export
Copy, download or export .pptx

Core Novelty

- ❑ One source → ***up to 10 formats*** in a single run.
- ❑ Compare outputs from ***up to 3 AI models*** side by side.
- ❑ Refine any draft in ***plain words***; no prompting skill needed.

Technical Moat

- ❑ ***Private mode:*** local LLM via Ollama (any locally installed model); data never leaves the device.
- ❑ Model gateway across ***Groq, Gemini & OpenRouter*** with retries.
- ❑ ***On-device OCR*** (Tesseract.js) for scanned pages.

"Seva First" Value

- ❑ Information reaches citizens in ***their own language*** and simple words.
- ❑ Small offices & NGOs get ***professional material*** without comms staff.
- ❑ Runs on ***free / low-cost models*** and ordinary hardware.

Proposed Solution Explanation / Demonstration Video

Video Link: https://youtu.be/u4sPNRXTmRE | **Live App:** https://sankshep-sfic.vercel.app | **GitHub:** SankShep.ai-SFIC

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TECHNICAL ARCHITECTURE || METHODOLOGY

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INPUTS

Documents
- ☐ PDF
- ☐ DOCX
- ☐ TXT · CSV · MD
- ☐ JSON

Scanned PDF & Images

Web Link
Cloud mode only

Pasted Text

Audience Profile
preset or saved

Processing Mode
Cloud or Private

STEP 1

Sign in
Supabase Auth

STEP 2

Processing Mode
Cloud APIs · Private
Private: checks Ollama + hardware, shows a setup guide

STEP 3

Configure Run
Source: upload · link · paste
Formats + instructions · tone · audience
Cloud: 1 model or compare up to 3
+ own API key (optional)

STEP 4

Plan the Run
Smart check against the canvas:
- only new formats are sent
- or the user regenerates all

STEP 5

Read Your File (in browser)
- PDF → text (pdf.js legacy build)
- DOCX → text (mammoth)
- TXT / CSV / JSON / MD → read directly
- Scanned PDF → 10 page images, ≤2000px

STEP 6

Links & Images → Text
- Link → page text via Jina Reader (Cloud only)
- Image → vision AI (Gemini, Mistral)
- If it fails → on-device OCR (Tesseract.js)
- Private mode: always OCR, images stay local

STEP 7

Clean Up Text
local tidy-up, no AI call
so nothing gets cut off

AI Orchestration · pipeline.ts (LangGraph-style)

STEP 8

Fit to Each AI
Groq 12k · Mistral 60k · Gemini 200k · on-device 6k chars
long docs sampled evenly

STEP 9a

Write the Drafts
one node per format × model
Cloud: in parallel · 2 × 3 = 6 drafts
Private: one at a time, streamed

STEP 9b

Translate (if asked)
whole text in small chunks
cut-off chunk → halved, redone

STEP 11b

Refine a Draft
type a change; AI picks draft-only or full source

STEP 10 · Model Gateway

- Own key → straight to the provider
- No key → our server's shared keys
- Private → Ollama on your computer
- Busy → retries · key fails → asks first

AI INFERENCE

Cloud · fast
Groq · Gemini · Mistral

On-device · private
Ollama · any local model

STEP 11a

Output Canvas
live, side by side
compare · export · Stop
.md · .txt · .pptx

STEP 12 – 13

Supabase
12 · Log the run
Private: counts only
13 · History
analytics & admin

Technology Stack

Frontend
React · TypeScript · Vite · Tailwind

Backend & Hosting
Node.js · Vercel

Database & Auth
Supabase · PostgreSQL

AI Inference
Groq · Gemini · Mistral · Ollama

Orchestration
LangGraph-style (TS)

Documents
pdf.js · mammoth · Tesseract.js · pptxgenjs

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TECHNOLOGY STACK & FEASIBILITY

Component Readiness

Layer	Tools	Status
Frontend	React 19, TypeScript, Vite 8, Tailwind CSS 4, React Router	✓ Built
Backend & API	Vercel serverless function /api/chat (holds the AI keys)	✓ Built
Database & Login	Supabase Auth + Postgres with row-level security	✓ Built
Cloud AI	Groq, Google Gemini, OpenRouter	✓ Integrated
Edge / Local AI	Ollama + any locally installed model on the user's own computer	✓ Integrated
Document Processing	pdf.js, mammoth, Tesseract.js OCR, Jina Reader, pptxgenjs	✓ Built
Translation	13 languages, incl. Hindi, Gujarati, Marathi, Bengali, Telugu, Tamil	✓ Built
Video Input	Transcription, keyframe & scene analysis	◷ Planned

Key Dependencies:- internet + AI API keys for Cloud mode; Ollama installed locally for Private mode (built-in hardware check & setup guide).

Feasibility

Technically:- Existing AI models, OCR & document tools make it buildable; the prototype is already live.

Economically:- Free / low-cost models, own-key option & hybrid AI keep costs low.

Operationally:- Simple upload-to-output flow with human review; works on any device.

Scalability:- Serverless backend + Supabase grow with users; task queues planned.

Proven:- Live demo at sankshep-sfic.vercel.app with a ready demo account.

Risks & Mitigation

- ❑ **API rate limits** → queuing & backup AI providers
- ❑ **Wrong AI output** → human review & refine before export
- ❑ **Very long documents** → chunking & even sampling per model
- ❑ **Weak hardware** → hardware check before Private mode
- ❑ **Sensitive files** → Private mode keeps them on-device
- ❑ **Many users** → background queues & auto-scaling

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SOCIO-ECONOMIC IMPACT & VIABILITY

1 → 10
output formats from one upload

13
languages supported

3
AI models compared side by side

2
AI modes: Cloud (fast) & Private (on-device)

For Citizens

- ❑ Alerts & schemes reach people ***faster***, in their ***own language***.
- ❑ ***Plain words*** instead of dense official language.
- ❑ Same info as ***posts, videos & infographics***.
- ❑ Better awareness → ***higher uptake*** of schemes.
- ❑ Clear, consistent info builds ***trust in public services***.

e.g. PHC health advisory → simple Gujarati explanation

For Govt. & Social-Sector Teams

- ❑ Hours of rewriting become ***minutes***.
- ❑ One upload → every version, ***consistent facts***.
- ❑ ***Private mode*** lets sensitive files use AI safely.
- ❑ Small offices & NGOs get ***professional material***.
- ❑ ***Human review*** before publishing keeps staff in control.

e.g. One circular → summary, slides, post & video script

Economic Impact

- ❑ Lower content-production & ***translation costs***.
- ❑ Staff time freed for ***field work & service delivery***.
- ❑ Runs on ***free-tier AI models*** & ordinary hardware.
- ❑ Affordable for ***district & panchayat offices***.
- ❑ Less need for paid ***translators & designers***.

e.g. Free Groq / Gemini models + own API key option

Viability

Target Users:- Govt. departments, district & panchayat offices, PHCs, NGOs and community organisations.

Adoption:- Simple upload-to-output web app, installable as a PWA; works on any device.

Sustainability:- Free-tier AI models, own-key option and hybrid cloud + on-device AI keep running costs low.

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IMPLEMENTATION ROADMAP & MILESTONES

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Development Phases

Where we are: *Phase 1 done & live → Phase 2 is next*

▼ WE ARE HERE

PHASE 1
Prototype Ready

Done ✓

- ❑ Full flow works: upload → AI drafts → download
- ❑ Compare up to 3 AI models
- ❑ Private mode: AI on your own computer
- ❑ Admin panel, history & analytics live

PHASE 2
Make It Reliable & Test

Month 1 – 2

- ❑ Keep working when free AI limits are hit (backup AI)
- ❑ Faster, cheaper replies by reusing past results
- ❑ Test with real users (students & faculty)
- ❑ Fix the problems they report

PHASE 3
Add Video Support

Month 3 – 4

- ❑ Turn speech in videos into text
- ❑ Pick key frames & understand each scene
- ❑ More output templates & download formats
- ❑ Better reading of scanned documents

PHASE 4
Grow & Launch

Month 5 – 6

- ❑ Handle big files in the background
- ❑ Server grows automatically with more users
- ❑ Security check & fair usage limits
- ❑ Bring in govt. offices, NGOs & colleges

Key Milestones

M1	M2	M3	M4	M5
Live prototype on Vercel	**Reliable AI with backups**	**First test with real users**	**Video support (beta)**	**Scaled public launch**
✓ Done	→ Next			

Success Metrics

- ❑ **Time saved** per document vs. manual rewriting
- ❑ **Generation success** rate (no failed or cut-off runs)
- ❑ **Output quality** from human review & refine
- ❑ **Active users** & formats generated (Analytics)

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MULTI-DISCIPLINARY TEAM & MENTORSHIP

Mentor



Khushali Mam

Faculty Mentor

VGEC (IT)

- **Department:** Information Technology
- **Mobile:** +91 97146 11388
- Guides ***problem framing, reviews*** & validates the solution.

Team Details

- **Challenge:** Seva First Innovation Challenge (SFIC) 2026
- **Team Code:** TM003665
- **Team Leader:** Siddharth Devmurari
- **Team Size:** 6 members + 1 mentor
- **Institute:** Vishwakarma Government Engineering College

Core Team & Ownership

Siddharth Devmurari

GenAI Systems & LLM Orchestration Lead

TEAM LEADER

Owns:-

System architecture & project planning, end-to-end GenAI pipeline and multi-LLM orchestration (Groq, Gemini, OpenRouter).

Suhani Jain

UI/UX Design Engineering & PWA Lead

Owns:-

Website UI/UX (Landing, Features, How It Works, About & Contact), design system & theme, responsive layouts and the installable PWA.

Harshit Vadher

Edge AI & Local LLM Inference

Owns:-

Privacy-first Private mode: on-device LLM via Ollama (any locally installed model), real-time streamed output, hardware compatibility check and guided setup.

Vraj Patel

Multilingual NLP & Document Intelligence

Owns:-

NLP-based translation across 13 languages, document ingestion & OCR (PDF, Word, scanned images) and PowerPoint (.pptx) export.

Kalindi Purohit

Frontend Architecture & Workspace Engineering

Owns:-

Core workspace UI: settings sidebar, live output canvas, side-by-side model comparison and Generate / Stop controls.

Kalee Hindocha

Backend Engineering & Data Security

Owns:-

Supabase database with row-level security, authentication (login & sign-up), protected routes, Admin panel, History and Analytics.